MPLS L3VPN & L2VPN Q&A – Arrcus Prep

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How to use: cover the answer, say yours out loud (control plane → data plane → failure → verify), then compare. ★ = areas where past Arrcus candidates failed: **who allocates which label**, control-plane procedures, PW status signalling, Nokia SDP, "knows config but not how it works".

Part A – MPLS foundations

Q1. Describe the MPLS label and the basic operations.

- 32-bit shim: **Label 20 bits**, **TC (EXP) 3 bits**, **S (bottom of stack) 1 bit**, **TTL 8 bits**.
- Operations: **push** (ingress LER), **swap** (LSR), **pop** (penultimate or egress).
- FEC = set of packets forwarded the same way (e.g. "to PE2 loopback /32"). LIB holds all bindings; LFIB holds the ones in use.
- Reserved labels: 0 IPv4 explicit-null, 1 router alert, 2 IPv6 explicit-null, 3 implicit-null, 13 GAL.

Q2 ★. How does LDP work?

- **Discovery:** hellos on UDP 646 to 224.0.0.2 (link) or unicast (targeted LDP, used for pseudowires).
- **Session:** TCP 646 between **transport addresses** (usually LDP router-ID / loopback) – must be reachable, else session stuck.
- Default modes (most vendors): **downstream unsolicited** (advertise labels without being asked), **liberal retention** (keep labels from all neighbours → fast reconvergence), **ordered or independent control**.
- Labels are advertised for IGP prefixes, by default /32 loopbacks (Cisco: all IGP prefixes).

> **Say it:** *LDP discovers neighbours with UDP 646 hellos, builds a TCP 646 session between transport addresses, and each router advertises its own local label for each FEC to all neighbours. Labels are locally significant and allocated by the downstream router.*

Q3 ★. What is PHP? Implicit-null vs explicit-null?

- Egress PE advertises **implicit-null (3)** → penultimate router **pops** the outer label, so the egress does only one lookup (VPN label or IP).
- **Explicit-null (0)**: penultimate swaps to label 0 → egress still gets a label with **TC/EXP** bits – preserves QoS marking end to end (pipe/short-pipe models).
- Without PHP: egress would pop the transport label then look up again – double work.

Q4. How does TTL behave and how do you hide the core from traceroute?

- Ingress copies IP TTL to label TTL (uniform model) → traceroute shows P routers.
- Disable TTL propagation (no mpls ip propagate-ttl / pipe model): label TTL = 255 → core hops hidden.
- P routers can still generate ICMP for VPN traffic by forwarding it along the LSP to the egress PE which routes it back.

Q5. LDP-IGP sync and session protection – why?

- **LDP-IGP sync:** when a link comes up, IGP converges before LDP labels exist → traffic sent unlabelled into the core → VPN black hole. Sync advertises max metric until LDP is up.
- **Session protection:** keeps the LDP session (via targeted hello) alive during a link flap → labels retained, faster recovery.

Part B – L3VPN control plane ★

Q6 ★. Who allocates each label in an L3VPN, and how is it signalled?

Label	Allocated by	Signalled by	Purpose
Transport (outer)	Each **downstream LSR** for the PE loopback FEC	LDP / RSVP-TE / SR (prefix SID + SRGB)	Reach the egress PE
VPN (inner)	**Egress PE**	**MP-BGP** VPNv4/VPNv6 (AFI 1/2, SAFI 128)	Identify VRF / CE / prefix at egress

> **Say it:** *Two labels from two control planes: the outer transport label is allocated hop by hop by each downstream router through LDP or SR, and the inner VPN label is allocated by the egress PE and advertised in MP-BGP with the VPNv4 route.*

Q7. VRF, RD and RT – define each crisply.

- **VRF:** separate routing table + FIB + interfaces per customer on a PE.
- **RD (8 bytes):** prepended to the IPv4 prefix to make **VPNv4 (RD:prefix)** unique when customers overlap. Formats: ASN:nn, IP:nn. **No role in import.**
- **RT (extended community):** policy – export RT attached on send; a VRF imports routes whose RT matches its import RT.

Q8 ★. Walk the control plane from CE1 to CE2.

1. CE1 advertises 10.1.1.0/24 to PE1 (eBGP/OSPF/static) into VRF A.
2. PE1 creates VPNv4 route **RD:10.1.1.0/24**, attaches **export RT**, allocates a **VPN label**, sets **next-hop = PE1 loopback**.
3. Sends via MP-iBGP (to RR) with send-community extended.
4. PE2 receives; imports into each VRF whose **import RT** matches; resolves next-hop PE1 via an **LSP** (needs a transport label for PE1 /32).
5. PE2 installs in VRF A FIB with [transport label, VPN label] and advertises to CE2.

Q9 ★. Walk the data plane from CE2 to CE1.

1. CE2 sends IP packet to PE2.
2. PE2 looks up in VRF A → pushes **VPN label** (inner) then **transport label** to PE1 (outer).
3. P routers **swap** only the outer label.
4. Penultimate P **pops** the outer label (PHP).
5. PE1 sees the VPN label → identifies VRF/CE → forwards IP to CE1 (or VRF lookup if per-VRF label).

Q10. VPN label allocation modes?

Mode	Labels	Egress behaviour
Per-prefix	One per prefix – most	Label → outgoing interface, no IP lookup
Per-CE / per-next-hop	One per CE	Label → CE adjacency
Per-VRF	One per VRF – fewest	Pop label then IP lookup in VRF (needed for aggregation)

Q11. Route reflector for VPNv4 – anything special?

- RR doesn't need VRFs; it keeps all VPNv4 routes (retain route-target all is implicit on RR). A normal PE **drops** VPNv4 routes whose RT it doesn't import.
- **RT-Constraint (RFC 4684, RTC):** PEs tell the RR which RTs they want → RR sends only those → scale.
- RR reflects best path per RD:prefix – hence the next question.

Q12. Same RD or unique RD for a dual-homed site?

- Same RD on both PEs: RR sees two paths for the same RD:prefix, picks one, reflects only the best → remote PEs know only one path → no ECMP, slower failover.
- **Unique RD per PE per VRF**: different VPNv4 prefixes → RR reflects both → remote PE imports both into the VRF → ECMP / BGP PIC.
- Alternative: ADD-PATH.

Q13. RT design examples – hub-and-spoke and extranet.

- **Full mesh:** all VRFs export and import RT 100:1.
- **Hub-and-spoke:** spokes export 100:SPOKE, import 100:HUB; hub imports 100:SPOKE and exports 100:HUB → spoke-to-spoke traffic goes via hub.
- **Extranet / shared services:** shared VRF exports 100:SHARED; customer VRFs import it (plus their own), and the shared VRF imports each customer RT.
- RT filtering scenario: use import/export route-maps to leak only specific prefixes.

Part C – PE-CE protocols

Q14 ★. Customer uses the same AS 65001 at two sites with eBGP to the PE. Site B doesn't learn Site A. Why?

- Route arrives with AS_PATH "100 65001" → CE sees its own AS → **eBGP loop prevention** drops it.
- Fix: **as-override** on PE (replace 65001 with 100) or **allowas-in** on CE.
- With multihomed sites add **SoO (Site of Origin)** extended community to prevent loops.

Q15. OSPF as PE-CE – what are the DN bit, domain-ID and sham-link?

- OSPF routes are carried across BGP with extended communities (route type, domain-ID). If domain-IDs match, the remote PE presents them as **Type 3 inter-area**; if not, as Type 5 external.
- **DN bit:** PE sets it in Type 3/5/7 LSAs sent to the CE; another PE receiving such an LSA **ignores** it – prevents re-injecting into BGP (loop).
- **Sham-link:** a link between PEs inside the VRF area so MPLS path appears **intra-area** – needed when a backdoor link between sites would otherwise always win (intra beats inter).

Part D – L3VPN troubleshooting

Q16 ★. CE1 cannot reach CE2. Your checklist?

1. **PE–CE:** CE route in VRF on PE1 (show ip route vrf A).
2. **MP-BGP:** VPNv4 session up; route exported with correct RD/RT and VPN label (show bgp vpnv4 unicast vrf A <prefix>).
3. **Import:** route present on PE2 and imported into VRF A (RT match; RR sending it; send-community extended).
4. **Next-hop / LSP:** PE2 has a label for PE1 /32 through every hop (show mpls forwarding, ping/traceroute mpls ipv4 PE1/32).
5. **Return path:** same checks in reverse.
6. **MTU:** large packets with extra 8+ bytes of labels.

Q17 ★. VPN routes look fine but traffic is black-holed in the core. Classic causes?

- PE loopback **summarised** in the IGP at an ABR → LDP has no label for the /32 FEC → LSP breaks; traffic arrives at a P router with only the VPN label, which it doesn't understand.
- PE loopback not /32 (e.g. /24) → label mismatch on some vendors.
- LDP not enabled on one core link; IGP uses it → unlabelled forwarding.
- LDP-IGP sync missing after link restore.
- Fix and verify with traceroute mpls (shows where labels stop).

Q18. MTU in MPLS?

- Each label adds 4 bytes: L3VPN = 8 bytes (2 labels), plus TI-LFA/RSVP backup labels, plus control word for PWs.
- Core MPLS MTU must be ≥ customer MTU + labels; otherwise large packets dropped (MPLS can't fragment easily) → symptoms: ping works, file transfer hangs.

Q19. Inter-AS options A, B, C.

Option	How	Pros / cons
A – back-to-back VRF	ASBRs connect with one sub-interface per VRF, plain IP between them	Simple, secure, per-VRF QoS; does not scale (VRF + session per customer)
B – VPNv4 between ASBRs	ASBRs exchange VPNv4 routes + labels via MP-eBGP; ASBRs hold all VPN routes	Scales better; ASBRs carry all VPN state
C – multihop MP-eBGP	RRs exchange VPNv4 directly; ASBRs exchange PE loopbacks + labels via BGP-LU	Most scalable; needs trust between providers (loopbacks leaked)

Part E – L2VPN: VPWS and VPLS ★

Q20. Compare VPWS, VPLS and EVPN.

	VPWS (E-Line)	VPLS (E-LAN)	EVPN
Topology	Point-to-point	Multipoint bridge	Both (EVPN-VPWS, E-LAN)
MAC learning	None	Data plane (flood-and-learn)	Control plane (BGP type 2)
Signalling	LDP (FEC 128/129) or BGP	LDP (RFC 4762) or BGP (RFC 4761)	MP-BGP EVPN
Multihoming	PW redundancy	Limited (active/standby)	All-active with ESI

Q21 ★. Describe a pseudowire's label stack and control word.

- Stack: [**transport label** → remote PE] [**PW label** → which PW/AC] [optional **control word**] [customer frame].
- PW label is exchanged over **targeted LDP** between PE loopbacks (or BGP for Kompella).
- **Control word (4 bytes):** first nibble 0 so P routers doing ECMP don't mistake the payload for IPv4 (4) or IPv6 (6) and mis-hash; also carries sequence number and flags. Both ends must agree.

Q22. LDP FEC 128 vs FEC 129? Martini vs Kompella?

- **FEC 128 (PWid):** each PE configured with the remote PE IP and a matching **VC-ID** – manual, common.
- **FEC 129 (generalised):** uses AGI/SAII/TAII identifiers; supports **auto-discovery** (BGP-AD) – used for VPLS.
- **Martini (RFC 4762, LDP):** full mesh of targeted LDP sessions for VPLS.
- **Kompella (RFC 4761, BGP):** BGP auto-discovery and label blocks; scales with RRs.

Q23 ★. Explain PW status signalling.

- Old method: when the AC fails, **withdraw the PW label** – slow, the whole PW must be re-signalled.
- RFC 4447: **PW Status TLV** sent in an LDP **Notification**; label stays installed, only status changes → faster recovery. Capability negotiated in the Label Mapping.

Code	Meaning
0x00	Forwarding
0x01	Pseudowire not forwarding
0x02	Local AC (ingress) receive fault

Code	Meaning
0x04	Local AC (egress) transmit fault
0x08	Local PSN-facing PW (ingress) receive fault
0x10	Local PSN-facing PW (egress) transmit fault
0x20	Standby (PW redundancy, RFC 6870)
0x40	Request switchover

> **Say it:** *Instead of withdrawing the PW label when the attachment circuit fails, the PE sends an LDP notification with a PW status TLV – for example 0x02 for a local AC receive fault. The label stays, the remote PE stops forwarding, and when the fault clears status returns to 0x00, so recovery is much faster.*

Q24. How does VPLS forward and prevent loops?

- Each PE runs a virtual switch per VPLS: learns MACs from ACs and from PWs (source MAC → PW).
- BUM is flooded to all ACs and all PWs.
- **Split horizon:** a frame received on a core (mesh) PW is never sent to another mesh PW – no STP needed, but a **full mesh** of PWs is required.
- **H-VPLS:** MTU-s connects to PE-rs with a **spoke PW** (no split horizon on spokes) → reduces mesh size.
- **MAC withdraw:** LDP Address Withdraw message tells remote PEs to flush MACs after a topology change (e.g. PW switchover).

Q25 ★. Nokia SR OS: explain SAP, SDP, mesh-SDP and spoke-SDP.

Term	Meaning
Service	VPLS / Epipe (VPWS) / VPRN (L3VPN) instance with a service ID
SAP	Service Access Point – customer attachment, e.g. 1/1/1:100 (port:VLAN)
SDP	Service Distribution Point – a logical **tunnel to a far-end PE** (far-end IP + transport LDP/RSVP/SR/GRE). Many services share one SDP
Mesh-SDP	Binding in a VPLS with **split horizon** among mesh-SDPs (full-mesh core)
Spoke-SDP	Binding **without** split horizon (H-VPLS, Epipe); carries a **VC-ID** that both ends must match

> **Say it:** *The SAP is where the customer enters, the SDP is the tunnel the service rides to the far-end PE, and the SDP binding with a VC-ID is the pseudowire. Mesh bindings get split horizon, spoke bindings don't.*

Q26. What is PW redundancy?

- Primary and backup PWs from the CE-facing PE to two remote PEs.
- Backup advertised with status **0x20 standby**; on primary failure, status flips and traffic moves; MAC withdraw flushes stale MACs.
- Test: fail the primary PE/link, measure loss, check status codes and MAC flush.

Part F – L2VPN troubleshooting and testing

Q27 ★. A pseudowire won't come up. Your checklist?

1. **Targeted LDP session** up to the remote loopback (show mpls ldp neighbor / show router ldp session).
2. **Transport LSP** to the remote PE exists.
3. **VC-ID** matches on both ends (FEC 128).
4. **MTU** matches – PW MTU mismatch keeps the PW down (LDP signals interface MTU).
5. **Encapsulation type** matches (Ethernet raw mode vs VLAN tagged mode, type 5 vs 4).
6. **Control word** setting agrees.

7. **AC status** – local SAP/port up; check PW status bits (local vs remote fault).

If asked "how would you test L3VPN/L2VPN?"

Area	Test ideas
Control plane	LDP/SR labels per FEC, VPNv4 export/import, RT filtering, RR + RTC, label modes, PW signalling (FEC 128/129), status TLV codes
Data plane	End-to-end label stack in captures, PHP vs explicit-null, ECMP, control word, MTU boundary, VLAN modes
PE-CE	eBGP (as-override, allowas-in, SoO), OSPF (DN bit, sham-link), static
Convergence	Core link/node failure, PE failure, PW redundancy switchover, LDP-IGP sync, BGP PIC – measured loss
Scale	VRFs, VPNv4 routes, PWs, VPLS MACs
Interop	Nokia ↔ Cisco/Juniper PW and VPN interop

Q28. Why is EVPN replacing VPLS?

- Control-plane MAC learning (less flooding), all-active multihoming with ESI, fast convergence (mass withdraw), integrated L2+L3 (IRB), one BGP control plane for L2/L3 and VPWS, works over MPLS, SR or VXLAN.